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**To cite this article:** Julio Galindo-Gutiérrez (2026) Banking against sustainable finance: the effect of the European central bank on green bonds, Review of International Political Economy, 33:2, 822-851, DOI: [10.1080/09692290.2025.2599249](https://doi.org/10.1080/09692290.2025.2599249)

**To link to this article:** <https://doi.org/10.1080/09692290.2025.2599249>



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Published online: 15 Dec 2025.



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# Banking against sustainable finance: the effect of the European central bank on green bonds

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## ABSTRACT

This paper highlights the unintended consequences of central bank practices on climate finance performance. By providing information on central banks' willingness to buy individual bonds from investors, collateral frameworks can reassure or spook market actors. While central bank collateral frameworks have been subjected to some analysis, their relationship with sustainable finance has been overlooked. By comparing matched green and conventional bonds, this paper finds robust empirical evidence of an anti-green bias at the European Central Bank (ECB), whose collateral framework systematically under-values green bonds relative to conventional bonds. It also shows that the ECB collateral framework moves markets; the lower value that the ECB imposes on green bonds translates into higher borrowing costs for green bond issuers, constraining the financing available to support the economic activities needed for a green transition. Given the ECB commitment to ensure that all of its policies are aligned with the objectives of the Paris Agreement, it is important that its impact on green bonds and potentially other forms of sustainable finance be subjected to further review and analysis.

**ARTICLE HISTORY** Received 30 January 2025; Accepted 21 November 2025

**KEYWORDS** Climate finance; central banks; green bonds; monetary policy, climate change

## Introduction

Climate change is one of the defining issues of our time, and we are at a crucial moment. The gravity of this issue led to the adoption of the Paris Agreement and underpins several of the Sustainable Development Goals (SDGs) agreed in 2015. Ever since, expanding the financing available to fund sustainable transitions has become a global priority (Agliardi & Agliardi, 2019, p. 608). It has been estimated that the world needs an

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 Supplemental data for this article can be accessed online at <https://doi.org/10.1080/09692290.2025.2599249>.

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annual investment of around 2.2% of global GDP to deliver the commitments from the SDGs and the Paris Agreement. Nonetheless, current levels of investment fall well short of this figure (GFANZ, 2023). In this sense, finding ways to unlock new and additional finance for climate-related projects is crucial. Private capital and market-based financing are seen as vital components of climate financing. In this regard, the development of green bonds has been called one of ‘the most important financial breakthroughs in the domain of sustainable finance during the last 15 years’ (Agliardi & Agliardi, 2019; Ando et al., 2022, p. 1). This innovative asset class is considered a principal mechanism through which to raise private capital to finance the fight against climate change (Bachelet et al., 2019; Langley et al., 2021).

Green bonds can unlock additional climate finance by trading at a premium, often called a ‘greenium’, which allows issuers to borrow at lower costs compared to similar conventional bonds. In this paper, I explore whether central bank behavior constrains or promotes this premium at which green bonds trade. Specifically, I assess the impact of the European Central Bank (ECB) collateral framework on green bond performance. Through its collateral framework, the ECB assesses the ‘riskiness’ of bonds and, as such, signals to market participants looking to use bonds as collateral for Central Bank credit the discount or haircut against prevailing market value they will apply when purchasing a bond. By using a matching method to identify essentially identical pairs of bonds, I find that as the ECB increases the discount it applies to green bonds, this increase results in less favorable borrowing conditions for issuers of green bonds than for issuers of identical conventional bonds. Furthermore, I find that the discount that the ECB applies to green bonds is substantially higher than the one it applies to conventional bonds.

In this paper, I engage with the broader scholarship on the political economy of central banking and climate finance (Abidi & Miquel-Flores, 2018; Andrade et al., 2016; De Santis, 2020; Eser et al., 2019; Galema & Lugo, 2021; Li et al., 2021; Pelizzon et al., 2023). Second, I draw on institutional research on how central bank design shapes political and economic outcomes (Ballard-Rosa et al., 2021; Beaulieu et al., 2012; Bodea & Hicks, 2015; Hix et al., 2010; Lombardi & Moschella, 2016; Mosley & Rosendorff, 2023). Third, and most significantly, I contribute to debates on the role of central banks in climate mitigation and adaptation, and to critiques of the limits of green central banking (Best et al., 2025; Boneva et al., 2022; Dafermos et al., 2021; Gabor & Braun, 2025; Kedward et al., 2024; Quorning, 2024; van ‘t Klooster & Fontan, 2020; van ‘t Klooster & Prodani, 2025). Theoretically, I introduce a distinction between two related but analytically distinct biases in central banking: a carbon bias and an anti-green bias. Both are allocative biases, but one refers to the

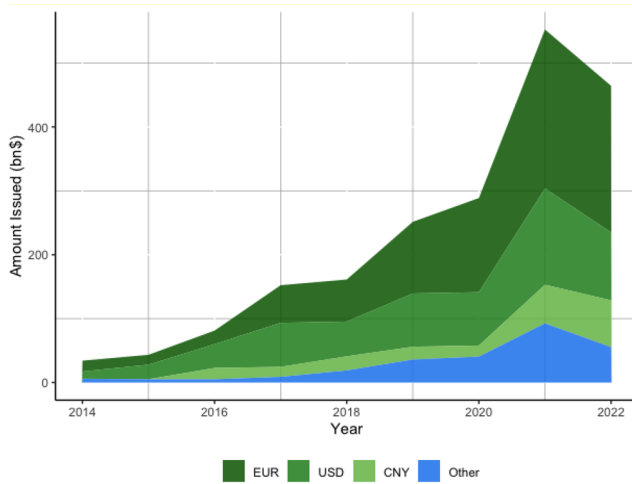
favorable treatment of carbon-intensive assets, and the other to an unfavorable treatment of green assets in haircut policy, all else being equal. I contribute to this debate by reframing haircut policy as a site where the structural contradiction between the ECB's formal mandate (price stability and market neutrality) and its informal climate commitments becomes visible and consequential. Furthermore, I bridge these macro-level critiques of green central banking that focus on market neutrality, weak derisking, and deferential riskification with observable price and allocation effects in financial markets.

The paper proceeds as follows. First, I introduce the empirical setting, explaining the nature of green bonds and the importance of green bonds trading at a premium to mobilize additional finance for green projects. Next, I theorize how the ECB's monetary policy interventions affect green bond pricing. Following that, I detail the variables, data sources, and methods used in my analysis. Finally, I present the results, highlight the theoretical and empirical contributions, and discuss implications for policy and future research.

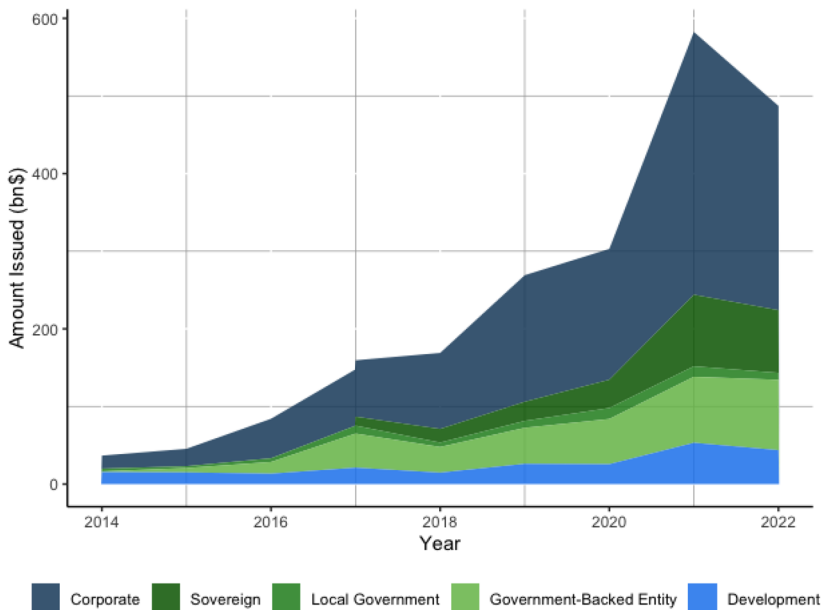
## **The green bond premium**

Green bonds refer to debt securities issued to raise capital. The structure of green bonds is essentially identical to that of conventional or 'plain vanilla' bonds. The structural similarity between green bonds and mainstream alternatives has led them to be labeled 'boring' (Bigger & Millington, 2020). The main difference between plain vanilla bonds and green bonds lies in the use of proceeds. The Green Bond Principles (GBP) of the International Capital Markets Association (ICMA) define green bonds as 'any type of bond instrument where the proceeds will be exclusively applied to finance or re-finance, in part or in full, new and/or existing eligible green projects' (ICMA, 2017). The second important difference between green bonds and conventional bonds is that, in addition to the standard credit rating process, green bonds can be issued with varying degrees of environmental assurance certification to avoid greenwashing (CBI, 2021; Jones et al., 2020; Simeth, 2022). Currently, green bonds are predominately denominated in Euros (EU€) (Figure 1), and corporate issuers have the largest share of the market (Figure 2).

While the size of the green bond market pales in comparison to the conventional bond market, other types of climate finance, such as the global carbon markets, the REDD carbon offsets, and the Green Climate Fund (GCF), appear as 'drops in the bucket of climate finance' compared to the green bond market (Bigger & Millington, 2020, p. 607).<sup>1</sup>



**Figure 1.** Green bond issuance by currency.



**Figure 2.** Green bond issuance by type of issuer.

### ***Green bond premium and additionality***

Actors who currently issue green bonds have little trouble raising capital—whether through green or conventional bonds (Jones et al., 2020). However, given that green bonds are touted as one of the solutions to address the need for funding climate change adaptation and mitigation

projects, it is important to question whether they can allocate funds toward environmental projects that would not otherwise be funded—in other words, whether or not they are scaling up additional sources of finance (Agliardi & Agliardi, 2019). This is the so-called question of the ‘additionality’ of green bonds. The concept of additionality for green bonds has a similar logic to the one used for the Clean Development Mechanism (CDM) (Michaelowa et al., 2009; Muller, 2009) and is defined as financing new projects that would have been impossible to fund through conventional bonds.

It is typically suggested that to achieve additionality, green bonds need to lower the cost of borrowing so that they can fund projects that are too expensive to finance through conventional bonds. For this reason, academic research on green bonds has focused, for the most part, on investigating the existence of a green bond premium or ‘greenium’ that investors are willing to pay for environmentally aligned debt (Cortellini & Panetta, 2021; MacAskill et al., 2021). The *greenium* is the difference in yield to maturity that a green bond has compared to an equivalent conventional bond. Thus, if a green bond has a lower yield to maturity than an equivalent bond, it is said to exhibit greenium (higher prices) because a higher price can be set. This implies that the borrowing costs the issuer faces are lower because investors are willing to accept lower yields in exchange for investing in a green project (Agliardi & Agliardi, 2019; Ando et al., 2022; MacAskill et al., 2021). The importance of green bonds having a ‘greenium’ would mean that environmentally aligned projects would be cheaper to fund *via* the debt markets than those from unlabeled debt, thus promoting additionality.

Theoretically, the green bond premium can either be positive or negative. On one hand, smaller issuances and less liquidity than conventional bonds can lead to positive premiums or lower prices for green bonds. On the other hand, the increase in demand for environmental, social, and governance (ESG) investments can lead to a greenium or lower prices for green bonds. In this sense, a greenium would provide new and additional climate financing as it would attract more capital at cheaper prices. However, evidence of the existence of a *greenium* is controversial (Ehlers & Packer, 2017; Flammer, 2021; Gianfrate & Peri, 2019; Kapraun et al., 2019; Karpf & Mandel, 2018; MacAskill et al., 2021).

### ***Determinants of the green bond premium***

Beyond whether green bonds provide cheaper funding than conventional bonds, the literature on green bonds seems to be preoccupied with establishing the determinants of such an effect. In particular, the literature has a narrow focus and discusses only internal financial characteristics of

bonds and environmental drivers that might lead investors to prefer one type of asset over the other, with no attention paid to some of the wider politico-economic factors that can influence bond prices, primarily the financial characteristics of the bonds such as their size, coupon rate, maturity, credit rating, currency of issue, and type of issuer (Bachelet et al., 2019; Febi et al., 2018; Simeth, 2022; Zerbib, 2019), as well as the effects of green label verification in the form of external reviews have on bond yields (Bachelet et al., 2019; Dorfleitner et al., 2022; Flammer, 2021; Simeth, 2022).

Apart from a few exceptions that address some external factors that might influence the green bond premium, such as tax incentives in United States (US) Municipal Bond market (Burton, 2018) or for corporate bonds in India (Agliardi & Agliardi, 2019), the financial economics literature is mostly focused on internal financial and environmental characteristics of green bonds, and pays little to no attention to wider politico-economic factors. By focusing on the role played by specific central bank policy interventions on green bond prices, I address this gap. Specifically, I probe the extent to which the ECB collateral framework and its valuation of haircuts influence green bond prices.

### **Theoretical framework: the role of the euro system collateral framework**

According to conventional wisdom, central banks are bastions of conservatism. However, this view may overlook their capacity for change and allow us to understand them as ‘unexpected revolutionaries’ (Moschella, 2024). In this sense, the issue of climate change has required that central banks reconfigure some of their core ideas in order to orient their decision-making process and operational practice to revise their inflation target frameworks to account for climate shocks (Boneva et al., 2022, pp. 773-777) and to alter their temporal parameters around decision-making (Thiemann et al., 2023). However, scholarship in institutional design and recent research in political economy have problematized the view that this transformation is either coherent or adequately ambitious. Rather, green central banking is emerging as a contested and constrained policy terrain (Best et al., 2025; DiLeo, 2023; Gabor & Braun, 2025). Understanding these limits is essential, given that since 2008 central banks have injected over US\$13 trillion (tn) into financial markets, a considerable scale that can entail deep implications for the trajectories of climate finance.

One key channel of central bank influence is the collateral framework, which determines how assets are valued and accepted in refinancing operations. Empirical studies have shown that more favorable collateral treatment improves borrowers’ financing conditions (Cassola & Koulischer, 2019;

Mésonnier et al., 2022; Nguyen, 2020; Nyborg, 2017; Pelizzon et al., 2023). Not all eligible assets are created equal; how central banks, and the ECB in particular, value eligible collateral affects bond yields generally and the greenium of green bonds in particular. The next subsection theorizes expectations concerning the effects of ECB collateral valuation on bond yields.

### ***The ECB valuation of collateral***

Collateral frameworks are an important mechanism through which central banks intervene in financial markets when a commercial bank requires a short term cash injection. Central banks rely on them to ensure that short term challenges facing an otherwise healthy institution do not spill over into broken contracts, a loss of confidence in that institution, or wider market distress. When a financial institution requires access to cash from a central bank, collateral framework operations allow the financial institution to sell an asset to the central bank in exchange for cash.

As a risk mitigation measure, the ECB needs to value the assets it accepts as collateral to ensure that the amount of central bank reserves borrowed against it does not exceed the actual value of the collateral in case the borrower defaults and the collateral it posted needs to be sold to recover the money lent (Bindseil et al., 2017). The essential risk mitigation tools that the ECB has at its disposal are marking-to-market on a constant basis, margin calls, and haircuts (Gabor & Ban, 2016; Pelizzon et al., 2023).

These risk mitigation tools require central bank staff to assess the perceived riskiness of assets. The most consequential of these tools is the use of haircuts when accepting collateral. In these instances, the ECB applies a discount, or haircut, in order to protect itself from a fall in its value. As such, a haircut establishes the amount of central bank reserves that a borrower receives in return for collateral and is defined as ‘the difference between the security’s price and the collateral value’ (Brunnermeier & Pedersen, 2009). For example, if a borrower posts as collateral an asset worth €100 that has a haircut of 5%, then the borrower would only receive €95 worth of ECB funds (Cassola & Koulischer, 2019). As such, haircuts are a risk mitigation tool designed to protect the ECB from a fall in the value of collateral in case it is forced to sell the assets at a lower price than the one it had when it was originally posted as collateral (ESRB, 2017).

The ECB estimates the percentage of the haircut for each security in its list of eligible assets based on the residual maturity of the asset (shorter maturities mean smaller haircuts), the type of asset (bonds, for instance, have lower haircuts than asset-backed securities), the type of



issuer (sovereign bonds are subject to smaller haircuts than corporate ones), coupon type, and credit rating (Adler et al., 2023; Barthélemy et al., 2018; Nguyen, 2020). However, the key overriding principle is that as the liquidity of an asset decreases, its haircut increases, as the goal is to calculate the amount of liquidity that can be backed by a particular asset in the case of default (Cassola & Koulischer, 2019; ECB, 2015).

Theoretically, given that assets with lower haircuts are inherently more attractive because they can be translated into central bank funds at a higher rate, lower haircuts would be translated into lower yields. Empirically, this has been tested by relatively few studies, although they all reach similar findings (Cassola & Koulischer, 2019; Nguyen, 2020; Nyborg, 2017; Nyborg & Woschitz, 2021). However, most analysis relies on a thin understanding of finance that treats haircuts as a purely technical risk-management tool. Yet, because they are set by the central bank, they are also a site of political and distributive choice (Paterson, 2021). As such it is important to investigate empirically the effect of haircuts on the greenium of green bonds, as well as expanding the theoretical space in which these analyses take place.

While these operational details appear purely technocratic, the criteria and levels set for haircuts reflect broader priorities and power relations in monetary governance. Because there is little empirical research on how collateral frameworks interact with green bond performance, it is necessary to situate haircut policy within the broader theoretical debates on green central banking (Best et al., 2025; Gabor & Braun, 2025; Kedward et al., 2024; van 't Klooster & Prodani, 2025).

For instance, Kedward et al. (2024) have argued that a structural consequence of the financial system where monetary policy and financial stability are ultimately dependent upon the 'investability' of bond markets is that central banks operate under a risk-based approach to policy where they accommodate the rise of market-based finance by fixing market signals, both by enhancing price discovery and by steering asset prices.

However, this risk-based approach is not conducive to climate goals because it treats climate risks as exogenous and to be priced in, rather than governed directly. As a result, tools like collateral haircuts reflect existing market biases and often penalize green assets as less liquid or riskier. Thus, I argue that the current risk-based model of central banking implicitly preserves existing allocative logics in financial markets. As such, haircut policies, even when presented as technocratic risk mitigants, may structurally disadvantage green assets by reflecting prevailing liquidity and credit market biases, thereby suppressing the greenium. Given these considerations, I theorize that ECB's haircuts negatively impact the potential greenium effect of green bonds

### ***Institutional design, market neutrality and haircuts***

In the preceding section I presented my hypothesis that the current risk-based approach of central banking would lead to haircuts suppressing the greenium effect in green bonds. Crucially, it is important to mention that central banks operate under a policy of market neutrality to ensure they do not distort the market where collateral trades; thus, we should not expect haircut policies to discriminate systematically against green assets. However, as I mentioned earlier, this conceptualization of collateral and haircuts relies on a thin understanding of finance that ignores the fact that attempting to mimic the market structure does not remove the political dimensions of asset purchases and offset unintended ‘distributive consequences’ (van ‘t Klooster & Fontan, 2020). In my case, I rely on an analytical distinction between carbon bias and anti-green bias. While similar, it is more useful to think of them as two sides of the same coin: One leads to the favorable treatment of carbon-intensive assets, and the other leads to the unfavorable treatment of green assets. Thus, to fully tease out the effects of these relationships, I build my theory on the anti-green bias that central banks have on green bonds by engaging in two distinct but related scholarship. First, political economy literature that problematizes the conception of ‘market neutrality’, and secondly the literature that points out the structural inconsistencies between central banks’ institutional design and this recent ‘ideational turn’ toward green central banking.

First, as I mentioned above, the ECB as well as other central banks operate under a policy of market neutrality. In theory, this entails that the ECB’s private sector asset purchases are done in proportion to how much they are worth in the market, and that loans against collateral do not have a substantive influence on the markets where collateral trades. For this reason, haircuts need to be estimated to appropriately reflect market size, liquidity, and credit risk over a ‘liquidation horizon, which is long enough to ensure it does not have a negative impact on prices through liquidation’ (Aubrechtova et al., 2023). In other words, market neutrality means that the ECB seeks to implement monetary policies in a way that does not give preferential treatment to specific sectors or asset classes within the financial markets to avoid distorting market dynamics (Bindseil et al., 2017).

However, the principle of market neutrality has been widely criticized in relation to the ECB’s climate ambition. For instance, given that the ECB’s asset purchases are done in proportion to the market index of the securities, this entails, in practice, a bias toward carbon-intensive sectors, as these tend to be capital-intensive (Boneva et al., 2022; Dafermos et al., 2021; Jourdan & Kalinowski, 2019; Matikainen et al., 2017; Monnin, 2018;

Papoutsis et al., 2021; van 't Klooster & Fontan, 2020). This bias lowers the borrowing costs in carbon-intensive sectors, and it has been argued that it leads to higher overall carbon emissions (Schoenmaker, 2021).

The ECB's adherence to market neutrality reinforces the effects of this risk-based approach, embedding existing market structures into collateral policy. For the most part, studies that have tested empirically this carbon bias have focused on the role that collateral eligibility criteria played during the ECB Corporate Sector Purchase Program (CSPP)—part of the broader Quantitative Easing program that the ECB conducted between June 2016 and December 2018—(Jourdan & Kalinowski, 2019; Matikainen et al., 2017; Papoutsis et al., 2021). Furthermore, the question of how haircuts help engrain this carbon bias in the framework has also been tested (Dafermos et al., 2021). As such, there is both theoretical and empirical evidence to suggest that central banks' policies have unintended 'distributive consequences' that negatively impact green assets.

This negative bias is not simply an unintended side-effect of policy design but reflects deeper tensions within the ECB's institutional architecture that make it ill-equipped to deal with the green bonds. Central banks are institutions and, as such, are composed of both formal and informal rules. Rational choice institutionalism argues that the alignment between formal rules and informal norms is a stronger predictor of economic success than the formal rules themselves. Misalignments can lead to inefficiencies, corruption, and stagnation (North, 1990, pp. 46-47). In this sense, if there is misalignment between a central bank's formal and informal rules regarding green bonds, it would negatively affect green bond performance.

Central banks face unique challenges in aligning their mandates with climate goals. Traditionally, the primary role of central banks has been to maintain price stability and financial stability. However, climate change poses risks to financial stability, prompting central banks to reconsider their roles in addressing the crisis (Bolton et al., 2020), and central banks have been leading financial policy discussions on climate finance ever since former Bank of England (BoE) Governor Mark Carney's landmark speech on climate-related financial risks in 2015.<sup>2</sup>

The ECB exemplifies the challenges of aligning formal and informal rules with enforcement mechanisms in the context of climate finance. Under Christine Lagarde's leadership, the ECB informally committed to intensifying its efforts to support the green transition. This includes incorporating climate considerations into corporate bond purchases, limiting the share of carbon-intensive assets that can be pledged as collateral, and promoting transparency to align with the Paris Agreement and European Union (EU) climate goals (ECB, 2022). However, these climate considerations remain informal and are not part of the ECB's formal mandate,

which prioritizes price stability through market neutrality. Market neutrality implies that monetary policy should not favor one sector over another. This formal principle creates tension with informal climate considerations, as prioritizing green assets could violate the neutrality principle.

This misalignment between formal and informal rules goes beyond a simple technocratic challenge. As recent scholarship in political economy has suggested, it reflects a deeper structural contradiction (Best et al., 2025; Gabor & Braun, 2025; Kedward et al., 2024; van 't Klooster & Prodan, 2025).

First, as I mentioned above, Kedward et al. (2024) have argued that central banks accommodate the rise of market-based finance by fixing market signals and derisking green assets, and because of this, there are limits in what they can achieve in greening financial flows, given that when decarbonization considerations are included in their collateral frameworks this can take the form of either 'carrots for green finance' or 'carrots and sticks' that rewards climate leaders and penalizes climate laggards (2024, p. 1603). However, this latter form would require central banks to suspend market neutrality. As such, Kedward et al. (2024) exemplify this by arguing that the ECB initially tilted its corporate bond purchases toward greener issuers (breaking with market neutrality) but soon reversed course under inflationary and political pressures. Thus, they have concluded that this failed experiment illustrates that the politics of inflation targeting (and by extension, of market neutrality) constitutes a significant obstacle to a green transition.

Adding to this line of argument, in their typology of green macro financial regimes, Gabor and Braun (2025) have classified the ECB as an example of a weak derisking regime in which climate policy is outsourced to market signals and private finance. However, if financial markets fail to adequately price green assets, the reliance on market neutrality will necessarily lead central banks to replicate existing financial risk models, which systematically disadvantage less capital-intensive, illiquid green assets. Thus, Gabor and Braun have argued that central banks are unable to move beyond this weak derisking model precisely because their mandates of inflation targeting stand in conflict with the demands of a stronger climate policy which requires that 'central banks give up market neutrality in favor of green credit policy' (2025, p. 553). Thus, the opposition between the ECB's formal (inflation targeting) and informal rules (climate policy) is not a mere issue of misalignment that can be addressed but a structural contradiction.

These regime-level dynamics align with wider political economy analyses showing how neoliberal governance norms constrain central bank climate action. Best et al. (2025) find that most Western central banks shaped by norms of inflation targeting, central bank independence, and

‘market neutrality’, have confined themselves to questions of banking supervision and risk management. As such, collateral frameworks that treat green bonds unfavorably reflect this neoliberal heritage and institutional conservatism; while some central banks recently adopted more ambitious climate strategies, the results so far have been mixed (Best et al., 2025).

At the operational level, on the other hand, these structural limits are expressed through what van ‘t Klooster and Prodani (2025) have termed ‘deferential riskification’—a banking supervisory mode in which climate-related financial risks are modeled and managed using backward-looking, market-based metrics. They have noted that even large banks struggle to quantify climate-related risks because uncertainty over government adaptation strategies makes forward-looking estimates difficult. Crucially, because this current regime of deferential riskification sustains market neutrality, if markets fail to price climate risk accurately, this approach can entrench mismatches in collateral treatment, reinforcing preferential financing for carbon-intensive assets and imposing relatively unfavorable financing conditions on green ones. On this basis, I expect that the ECB’s collateral framework will systematically assign higher haircuts to green assets.

Taken together, these insights sketch my theoretical expectations about how the ECB’s collateral framework shapes financing conditions for green bonds. Hypothesis 1 tests the *price effect* of haircuts, Hypothesis 2 tests the *allocative bias* or anti-green bias of the Central Bank.

**H1:** Higher haircuts in the ECB collateral framework suppress the greenium effect for green bonds.

**H2:** The ECB collateral framework systematically treats labeled green bonds less favorably than comparable conventional bonds.

In the following sections, I explain the empirical setting of green bonds and the green bond premium, as well as the research design I follow to test these hypotheses.

## Research design

### *Outcome and predictive variables*

The outcomes of interest are variations of bond yields. For H1, I rely on the bonds yield to maturity (or the anticipated return of a bond if held until maturity). For H2, the main outcome of interest is the green bond premium or greenium. I estimate the greenium as a yield spread, that is, the difference in yields to maturity of a green bond and a conventional bond. Conventionally, yield spreads are estimated by deducting the yield

of one bond from the other and this difference is expressed in basis points (bps).<sup>3</sup> As such, I estimate greenium as the difference between the daily green bond yield to maturity (*mid*) and the one from a similar conventional bond according to the following equation:

$$\Delta y_{i,t}^x = y_{i,t}^{gb} - y_{i,t}^{cb} \quad (1)$$

where  $\Delta y_{i,t}$  is the greenium for the  $i$ th bond couple on day  $t$ ,  $y_{i,t}^{gb}$  is the yield to maturity of green bond  $i$  on day  $t$ , and  $y_{i,t}^{cb}$  is the yield to maturity of a similar conventional bond  $i$  on day  $t$ .

As mentioned, the green bond premium could take either a positive sign or a negative sign. However, to achieve additionality, the greenium should take a negative sign in order to be able to fund climate-aligned projects more cheaply than conventional bonds.

My main explanatory variables are the green bond label and the haircut spread. First, the green bond label is a dummy variable coded 1 if the bond is labeled green and 0 otherwise. Secondly, the haircut spread is estimated similarly to the yield spread as the difference in haircut of a green bond and that of a similar conventional bond on the same trading day:

$$\Delta \text{Haircut}_{i,t}^x = \text{Haircut}_{j,t}^{gb} - \text{Haircut}_{k,t}^{cb} \quad (2)$$

where  $\text{Haircut}_{i,t}^{gb}$  is the value of the haircut that the ECB assigns to green bond  $j$  of pair  $i$  on day  $t$ , and  $\text{Haircut}_{k,t}^{cb}$  is the yield to maturity of a similar conventional bond  $k$  of pair  $i$  on day  $t$ .

### **Data sources and empirical strategy**

The main source of data is the Refinitiv database. In addition to this, I use the ECB general collateral framework eligible assets database. First, I used the Refinitiv Government and Corporate Bonds Advanced Search application from which I extracted an initial sample of all active green and conventional bonds issued between October 2017—when sovereign green bonds were first issued—and May 2023.

To avoid the uncertainty of the impact that a non-constant coupon rating can have on the pricing at issuance and to construct a more homogeneous sample, I excluded bonds without fixed coupons—as they have non-constant coupons linked to a particular reference rate—and focus exclusively on plain vanilla bonds to control for the uncertainty that floating rates can have on prices at issuance (Gianfrate & Peri, 2019; Simeth, 2022). I then screened for duplicate bonds and those without International Securities Identification Number (ISIN) and removed them from the

sample. To control for the impact that a lack of credit rating might have on a bond's price, I also excluded bonds without ratings. I relied on ratings provided by Moody's as it had the highest number of observations in the data set and complemented them with Fitch ratings when available.

Then, for each green and conventional bond, I collected descriptive information such as issuer, issuer type, and currency. Furthermore, I collected several structural parameters to use as control variables such as coupon rate, date of maturity, original yield maturity to account for the pricing at issuance of the bonds, and amount issued given that substantial differences of these variables between bonds can have a significant effect on their yields (Zerbib, 2019). Finally, I collected daily observations of the mid yields as well as daily observations of the ask and bid prices of each bond, as the bid-ask spread is a common proxy for estimating a bond's liquidity where a higher spread is associated with lower liquidity (Adrian & Shin, 2010; Bachelet et al., 2019; Febi et al., 2018). Bonds with missing values were excluded from the sample. The resulting database consists of 28,137 bonds of which 999 are labeled green.

I merged this database with data from the ECB eligible asset database, which lists all eligible marketable assets and is updated on a daily basis. Specifically, for each bond I extracted, whether it was included in the framework or not, and, for bonds that did meet the eligibility criteria, I also extracted daily haircut observations. Table 1 summarizes the variables I used and their descriptions.

A common strategy for estimating the treatment effects with panel data is with two-way fixed effects regressions. However, the difference between the amount of conventional bonds versus green bonds leads to an imbalance in covariates which, in turn, would lead to inefficient estimations of causal effects and higher degrees of model dependence (Imai et al., 2023; King & Nielsen, 2019). Moreover, to effectively estimate the treatment effect of the green label on a bond, it would be necessary to compare the outcome variable of a bond had it been designated as green and of the

**Table 1.** Description of variables.

| Variable  | Description                                                 |
|-----------|-------------------------------------------------------------|
| ISIN      | International securities identification number for the bond |
| Issuer    | Name of issuer                                              |
| Amount    | Amount issued in US dollars                                 |
| Coupon    | Coupon rate of the bonds                                    |
| Maturity  | Maturity of a bond in years                                 |
| Currency  | Currency of amount issued                                   |
| Rating    | Credit rating (Aaa = 21; BBB = 12)                          |
| Pask      | Ask price                                                   |
| Pbid      | Bid price                                                   |
| Liquidity | Bid-Ask spread                                              |
| Yield     | Mid yield of a particular bond                              |
| Haircut   | Haircut percentage that the ECB assigns to a specific bond  |

same bond if it had been issued conventionally. However, this analytic approach requires that we observe each bond as being priced as both green and conventional, which is not possible. Finally, practical barriers prevent us from collecting the data through experimental design (Bachelet et al., 2019; Gianfrate & Peri, 2019).

In order to solve these issues, matching methods are a common tool for improving the validity of causal studies where counterfactual analysis or experiments are not possible by way of finding ‘hidden randomized experiments’ within an observational dataset. As such, they have become increasingly popular for comparing the returns of green bonds *vis-à-vis* conventional bonds (Bachelet et al., 2019; Gianfrate & Peri, 2019; Hyun et al., 2021; Simeth, 2022; Zerbib, 2019).

In essence, matching preprocesses data by ‘pruning’ observations that have no close matches on covariates in both the control and treated groups so that estimating the average treatment effect on the treated (ATT) is less of a function of modeling decisions, that is, less model dependent. Also, by creating pairs of observations that have essentially the same properties except for the treatment variable, it avoids the issue of selection bias. Finally, by removing heterogeneity, such models are more efficient estimators of the ATT (Iacus et al., 2012; Imai et al., 2023).

To this end, first I matched each green bond in the dataset with a conventional bond to its nearest neighbor in terms of selected characteristics. I specified the matching requirements as follows to control for individual (Table 2) characteristics and the credit risk of each bond. Zerbib has tested the robustness of these parameters and found not only that they are robust and efficient in balancing the covariates of the observations, but also that more stringent matching constraints degrade the quality of the post-matching estimation (Zerbib, 2019, pp. 49-50). As such, these parameters have become widely accepted in subsequent analyses of bond prices (Bachelet et al., 2019; Simeth, 2022). Moreover, in order to control for any bias induced by potential mismatches, I controlled for variables that are not exactly matched in the subsequent econometric analysis. Despite the strictness of these parameters, I identified 265 pairs of green and conventional bonds in which the only

**Table 2.** Bond matching characteristics.

| Bond characteristic | Matching parameters |
|---------------------|---------------------|
| Amount issued       | ±400%               |
| Coupon              | ±0.25%              |
| Issue Date          | ±4 years            |
| Maturity            | ±4 years            |
| Currency            | Same                |
| Issuer              | Same                |
| Credit rating       | Same                |



**Table 3.** Descriptive statistics.

| Green Bonds        | N      | Mean     | St. Dev. | Min   | 25th Perc. | 75th Perc. | Max      |
|--------------------|--------|----------|----------|-------|------------|------------|----------|
| Yield              | 43,565 | 1.95     | 1.63     | −0.86 | 0.32       | 3.32       | 6.95     |
| Haircut            | 43,565 | 9.52     | 6.85     | 0.90  | 3.60       | 14.90      | 30.50    |
| Liquidity          | 43,565 | 0.35     | 0.27     | −1.02 | 0.16       | 0.44       | 3.72     |
| Coupon             | 43,565 | 0.90     | 1.01     | 0     | 0.01       | 1.2        | 5        |
| Size (\$ MM)       | 43,565 | 2149.85  | 3728.48  | 11.23 | 561.35     | 1122.70    | 17615.58 |
| Maturity           | 43,565 | 9.04     | 4.63     | 2.00  | 5.42       | 10.01      | 30.02    |
| Days to Maturity   | 43,565 | 2,737.67 | 1,702.11 | 61    | 1,540      | 3,404      | 10,573   |
| Conventional Bonds | N      | Mean     | St. Dev. | Min   | 25th Perc. | 75th Perc. | Max      |
| Yield              | 43,565 | 1.97     | 1.63     | −0.83 | 0.35       | 3.34       | 7.08     |
| Haircut            | 43,565 | 8.83     | 7.24     | 0.00  | 3.00       | 13.20      | 31.50    |
| Liquidity          | 43,565 | 0.34     | 0.29     | −1.99 | 0.15       | 0.43       | 4.96     |
| Coupon             | 43,565 | 0.87     | 1.01     | 0.00  | 0.01       | 1.30       | 5.25     |
| Size (\$ MM)       | 43,565 | 3320.19  | 7290.78  | 11.23 | 561.35     | 1295.48    | 31435.60 |
| Maturity           | 43,565 | 9.08     | 4.62     | 2.00  | 5.25       | 10.51      | 30.96    |
| Days to Maturity   | 43,565 | 2,647.91 | 1,704.94 | 32    | 1,469      | 3,297      | 11,304   |

significant difference is the green label. The matching process resulted in much more balanced covariates. To test the hypotheses I had theorized, I further narrowed the database to include only those bonds included in the ECSF and with haircuts available. This narrowed the sample to 111 pairs of bonds (Table 3).<sup>4</sup>

The main dependent variable in Hypothesis 1 is the greenium effect or green bond premium that a green bond shows. I estimated the greenium as the daily yield spread between a green bond and its matched conventional bond as defined in Equation 1. As mentioned previously, this green bond premium can be negative (*greenium*) or positive (higher yields). To test my first hypothesis, I used the following model:

$$\Delta y_{i,t}^s = \beta_0 + \beta_1 \Delta Haircut_{i,t}^a + \beta_1 \Delta Liq_{i,t}^a + \sum_i \Delta \beta_i^a + \gamma_n + \eta_y + \epsilon_{i,t} \quad (3)$$

Here,  $\Delta y_{i,t}^s$  is greenium defined as in Equation 1. Thus,  $\Delta Haircut_{i,t}^a$  is my main explanatory variable and is defined as the haircut spread of a matched bond couple  $i$  on trading day  $t$  as in Equation 2. Conversely,  $\Delta Liq_{i,t}^a$  is the liquidity spread of a matched bond couple  $i$  on trading day  $t$  and the same for the control variables included in  $\sum_i \Delta \beta_i^a$ , which includes the set of bond characteristics that were not matched exactly such as issue amount, maturity, coupon, and credit rating. These controls are used throughout the financial literature on bond pricing (Bachelet et al., 2019; Kapraun et al., 2019; Simeth, 2022; Zerbib, 2019). I also included issuer ( $\gamma_n$ ) and yearly ( $\eta_y$ ) fixed effects to control for any unobserved heterogeneity. Finally,  $\epsilon_{i,t}$  is the stochastic error term clustered at the individual bond level.

To test the second hypothesis, the dependent variable is the haircut that the ECB applies to each bond in its collateral framework with the following model:

$$Haircut_{i,t}^s = \beta_0 + \beta_1 Green Bond_{i,t}^a + \sum_i \beta_i^a + \gamma_u + \eta_y + \epsilon_{i,t} \quad (4)$$

where  $Haircut_{i,t}^s$ , as mentioned, is the haircut that the ECB applies to each bond on each trading date. The quantity of interest is  $Green Bond_{i,t}^a$ , a dummy variable indicating whether the bond is labeled green or not, in order to test for the effect that being labeled green has on the ECB's valuation of haircuts. Finally,  $\sum_i \beta_i^a$  is a set of control variables to account for the other factors that the ECB uses to determine the degree of haircut value for each security. According to Article 2 of the Haircut Guideline of the ECB, the percentage of the haircut is determined by the type of issuer, the residual maturity, the coupon structure, the credit quality, and the liquidity category of the asset (Adler et al., 2023; Nguyen, 2020). Given that the sample of bonds consists only of plain vanilla bonds with fixed coupons, I controlled for differences in liquidity, residual maturity, and rating.<sup>5</sup> Finally, for completeness and robustness, I leveraged the fact that matching allows us to compare the effect of the treatment variable with a simple difference in means by testing these differences in both the haircut value and haircut spread between green and conventional bonds.

In addition to this, I ran several robustness checks to test the soundness of my research design. Specifically, I tested alternative specifications of the liquidity, relying on the ECB liquidity categories instead of the Bid-Ask Spread (Appendix 2.B). To address questions of whether this bias might change over time if it remains constant, I split the sample in two time periods—for bonds prior to the inflation crisis in 2022 and those after the inflationary crisis after 2022 (which roughly correlate with pre and post Covid time periods)—and found similar results to those in the main text, leading to the conclusion that this bias is persistent over time. Finally, to address concerns regarding the quality of the matching parameters, I followed Zerbib (2019) in further restricting the parameters and found the results to remain robust. These results are in line with Zerbib's own conclusions that in addition to restraining the obtained sample, requiring very stringent matching constraints can degrade the quality of the estimation.

## Results and discussion

I begin by showing in Table 4 the results from Equation (3), which presents the effects that ECB haircuts have on the green bond premium

(i.e., greenium) defined as the yield spread between a green bond and a matched conventional bond.<sup>6</sup> Higher haircuts lead to a negative greenium effect, as they result in green bonds having yield spreads that are 2.5 bps higher than those of conventional bonds. This offers statistically significant support for Hypothesis 1, which posits that higher haircuts negatively impact the green bond premium.

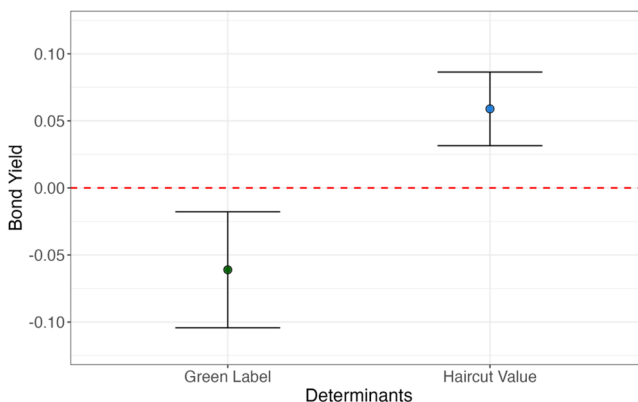
Looking at the effect that haircuts have on yields (as opposed to the effect that haircut spreads have on the green bond premium) reveals an interesting dynamic, as the effect of the haircuts offsets the potential effect of the green label. Figure 3 shows that, while being labeled green leads to bond yields that are 6.1 bps lower (holding all other variables constant), an increase in haircuts is associated with yields that are 5.9 bps higher. Thus, larger haircuts almost entirely minimize the greenium effect of green bonds, limiting the potential for additionality that such bonds can achieve. Conversely, the green bond label seems to offset the effect that higher haircuts have on the greenium effect. This explains why,

**Table 4.** Haircut effects on green bond premium.

|               | <i>Green Bond Premium (Greenium)</i> |                     |                     |
|---------------|--------------------------------------|---------------------|---------------------|
|               | (1)                                  | (2)                 | (3)                 |
| Haircut       | 0.025***<br>(0.008)                  | 0.047***<br>(0.011) | 0.047***<br>(0.011) |
| Issuer FE     | ✓                                    | ✓                   | ✓                   |
| Year FE       | ✓                                    | ✓                   | ✓                   |
| Liquidity     | ✓                                    | ✓                   | ∈                   |
| Full controls | ✓                                    | ∈                   | ∈                   |
| Observations  | 43,565                               | 43,565              | 43,565              |
| Bonds         | 111                                  | 111                 | 111                 |
| R2            | 0.315                                | 0.166               | 0.166               |

Note: Cluster robust standard errors in parentheses.

\* $p < 0.1$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .



**Figure 3.** Marginal effects on bond yields with 95% confidence intervals.<sup>8</sup>

despite having larger haircuts, green bonds have still been found to trade at lower costs in the literature.

Finally, Table 5 shows the results of Equation (4) and illustrates the net effect that the green bond label has on the haircut that the ECB assigns to a green bond, compared to the haircut the ECB assigns to an identical conventional bond. It shows that green bonds have a significant negative haircut premium, as the green label increases the spread by 66 bps. It is important to note that all models control for differences in liquidity, ensuring that the difference in yields and haircuts cannot be explained by green bonds showing lower levels of liquidity.<sup>7</sup>

For robustness, Figures 4 and 5 display the differences in means in haircuts (Figure 4) and haircut spreads (Figure 5) between green and conventional bonds. Relying on a two-sample t-test, the results reveal a statistically significant difference in means ( $p < 0.001$ ), with green bonds having a higher mean haircut ( $M = 9.52$ ) compared to conventional bonds ( $M = 8.83$ ), and a higher mean haircut spread for green bonds ( $M = 0.1$ )

Table 5. Effects on haircuts.

|               | Haircut Value      |                    |                    |
|---------------|--------------------|--------------------|--------------------|
|               | (1)                | (2)                | (3)                |
| Green Bond    | 0.664**<br>(0.285) | 0.661**<br>(0.283) | 0.687**<br>(0.309) |
| Issuer FE     | ✓                  | ✓                  | ✓                  |
| Year FE       | ✓                  | ✓                  | ✓                  |
| Liquidity     | ✓                  | ✓                  | ∈                  |
| Full controls | ✓                  | ∈                  | ∈                  |
| Observations  | 87,130             | 87,130             | 87,130             |
| Bonds         | 222                | 222                | 222                |
| R2            | 0.21               | 0.06               | 0.02               |

Note: Cluster robust standard errors in parentheses.

\* $p < 0.1$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

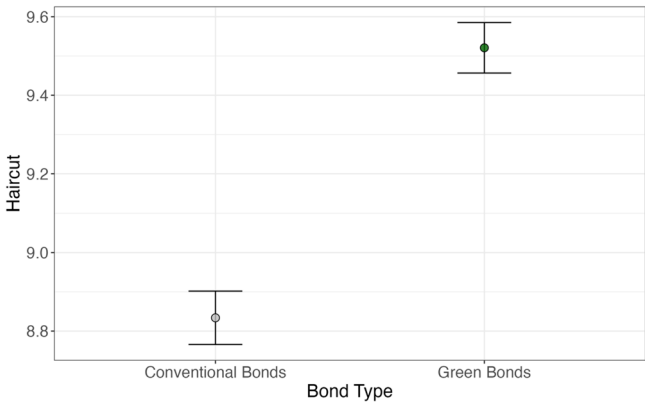
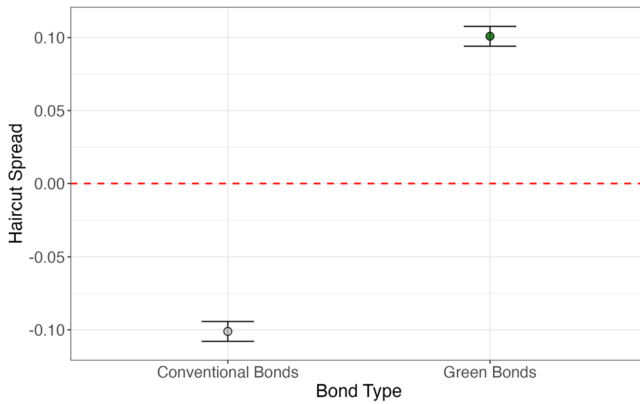


Figure 4. Difference in means of haircut values.



**Figure 5.** Difference in means of haircut spreads.

compared to conventional bonds ( $M = -0.1$ ). Taken together, these findings provide significant support for Hypothesis 2.

All the results I present are consistent with previous theoretical and empirical research on the fact that the ECB's principle of market neutrality leads to a bias in favor of carbon-intensive sectors. It is argued that central banks need to move beyond a strict application of market neutrality in order to support climate targets. As such, there have been several proposals on how the ECB could green its collateral framework by addressing this carbon bias (Dafermos et al., 2021; Jourdan & Kalinowski, 2019; Monnin, 2018).

An important theoretical step evidenced by my results is to distinguish between two closely related but not entirely overlapping concepts: carbon bias and anti-green bias. While similar, it is more useful to think of them as two sides of the same coin: One leads to the favorable treatment of carbon-intensive assets, and the other leads to the unfavorable treatment of green assets. Thus, to green its collateral framework, the ECB would need to address both its carbon bias and its anti-green bias.

As a corollary, the results of Hypotheses 1 and 2 support the theoretical distinction between carbon bias and anti-green bias. The ECB is contributing to green bonds having a less significant premium relative to conventional bonds, but this is not by supporting bonds issued in carbon-intensive sectors but by systematically treating green bonds less favorably through more conservative haircut valuations. In this sense, my empirical results suggest that ECB policy aligns more closely with what I describe as a 'sticks without sticks' approach, rather than the 'carrots without sticks' suggested in previous arguments (Kedward et al., 2024). That is, the ECB neither rewards green assets nor penalizes carbon intensive ones (no sticks), yet, it still imposes harsher treatment on green bonds (sticks).

I find robust evidence that the ECB exhibits an anti-green bias by valuing the haircuts of green bonds demonstrably higher than those of conventional bonds included in its collateral framework, as theorized in Hypothesis 2. This, in turn, leads to a negative greenium effect by increasing bond yields, as expected by Hypothesis 1. Therefore, to effectively green its collateral framework, the ECB would not only need to replace carbon-intensive bonds but also address the unfavorable treatment of green bonds that results in higher borrowing costs. It could do this by targeting green bond prices and addressing the higher discounts that green bonds carry relative to conventional bonds in its collateral framework.

As my results show, haircuts have a significant impact on the green bond premium, as a 1% difference in the haircut spreads leads to a 2.5 basis point difference in bond yields, making borrowing through green debt more expensive. As mentioned previously, one of the selling points of green bonds is that they could attract additional finance to fund green projects by lowering the cost of borrowing. Thus, by properly valuing green assets equally to similar conventional bonds when assessing haircuts, the ECB can help further lower the financing costs of additional green projects. This would help jumpstart the green bond market in particular, as well as other types of environmental assets, such as ESG bonds or sustainability-linked bonds, especially since the demand for these types of assets has seemed to decline since 2023 (Bryan, 2023; Financial Times, 2023; Webb, 2023).

These results lend empirical weight to recent work questioning the efficacy of risk-based, market-neutral approaches to climate finance. As it has been argued previously and extensively (Best et al., 2025; Gabor & Braun, 2025; Kedward et al., 2024; van 't Klooster & Fontan, 2020; van 't Klooster & Prodani, 2025), such strategies inadvertently reinforce financial structures that allocate credit away from green investments, in particular, by not penalizing brown assets given the carbon bias implicit in the principle of market neutrality. My results show that the haircut premium imposed on green bonds by the ECB exemplifies this tension, where climate-aligned assets are penalized under a regime nominally committed to climate goals.

My findings, however, also point to a crucial difference with previous research that typically advocates moving beyond risk-based, market-neutral approaches to climate policy making by central banks for more stringent reforms such as repoliticizing the role of central banks by challenging the traditional norms of central bank independence and market neutrality (Best et al., 2025; Kedward et al., 2024; van 't Klooster & Fontan, 2020), moving from a derisking status quo to a 'big green state' or a model alignment-focused supervision, as well as reducing the structural power of

finance (Gabor & Braun, 2025; van 't Klooster & Prodani, 2025). While, I agree with their conclusions that the misalignment between formal and informal rules that leads to these suboptimal outcomes seems to be structural given the contradictory nature between market neutrality and effective climate policy as theorized above, my results paint a more positive picture given that they reveal that within the ECB's current policy space, there are still simple technocratic fixes that can be applied in order to minimize the impact of this misalignment without compromising their mandates and central bank independence.

As such, in contrast to research that advocates for central bank collateral frameworks to actively incentivize green investments and penalize carbon-intensive activities, I argue for a more conservative approach. As I have demonstrated, the ECB exhibits an anti-green bias in its collateral framework, with green bonds facing haircuts approximately 60 basis points higher than those of conventional bonds. Therefore, addressing this anti-green bias in the ECB's collateral framework could significantly increase the green bond premium and promote environmentally aligned projects, without requiring aggressive policy changes that might undermine its commitment to market neutrality given that it is a contested policy terrain that can hinder 'non-reformist reforms'.

The ECB itself has periodically acknowledged these tensions, revealing both an awareness of the problem and the fragility of reform efforts. The urgency of the climate crisis has pushed the ECB to recognize the need to move beyond a strict application of market neutrality. For example, Isabel Schnabel, a member of the ECB's executive committee, acknowledged that the current bias supports a 'market structure that hampers an efficient allocation of resources' (Schnabel, 2021). In response, the ECB has pledged to incorporate climate change considerations into its monetary policy operations, including limiting the share of assets issued by carbon-intensive firms that can be pledged as collateral, and ensuring that all its policies align with the objectives of the Paris Agreement. However, when inflationary pressures resurfaced, the ECB scaled back these commitments and abandoned plans to decarbonize its bond portfolio, illustrating how even modest departures from market neutrality can become politically challenging.

## Conclusions

Existing scholarship has underscored the critical role bonds play in the economy. Over the past several decades, particularly since the global financial crisis, bond markets have become central to the financial system, with governments and corporations relying on them to access funding. In fact, bond markets have been described as the engine driving the global economy (Wigglesworth, 2023).

Theoretically, I distinguish anti-green bias in the ECB collateral framework as a political economy dynamic distinct from carbon bias. While carbon bias leads to the favorable treatment of carbon-intensive assets, anti-green bias leads to the unfavorable treatment of green assets. My findings provide robust evidence supporting my hypotheses, demonstrating the direct influence of the ECB on the greenium effect (H1) through its haircut valuation (H2). Green bonds are assigned significantly higher haircuts compared to conventional bonds, with a difference of approximately 66 basis points. These more conservative haircuts have a measurable impact, increasing the yield spread between green bonds and their conventional counterparts by around 2.5 basis points. This, in turn, negatively affects the greenium of green bonds, undermining their potential to attract additional financing for environmentally aligned projects.

I contribute to the emerging literature on green bond pricing by examining the impact of state agencies, particularly central banks, on bond prices. Traditionally, the focus has been on the intrinsic characteristics of bonds, such as maturity, sectoral positioning, and credit ratings, in determining the greenium. By shifting the focus to central bank practices, I highlight how institutions play a critical role in governing and influencing the greenium. This contribution extends to the broader literature on the political economy of climate change, demonstrating that some of the most influential climate policies are not explicitly climate-related. Specifically, I show how central banks, in their capacity as market makers, can shape investor incentives in ways that are misaligned with the goals of the Paris Agreement. Theoretically, I draw on the distinction between the ECB's carbon bias and anti-green bias within its monetary policy. To effectively green its policy, as it intends, the ECB must not only penalize assets from carbon-intensive sectors but also address the bias against those from green sectors.

My findings raise several important questions that warrant further investigation. First, the causal mechanisms behind the more conservative haircuts applied to green bonds remain unclear, indicating a need for further research into the underlying rationale for this discrepancy such as conservative risk-weighting methodologies or liquidity assumptions biased against emerging asset classes. Addressing this issue would require looking inside the ECB's decision-making 'black box' and process-tracing the causal chain leading to the bias, a task that calls for a fundamentally different approach than the one taken in this paper. While such an inquiry lies beyond the current scope, it opens an important avenue for future research. Second, the role of other state agencies in influencing green bond pricing deserves attention. For example, the EU's long-standing reputation as a climate leader (Biedenkopf et al., 2022) could play a significant role. In 2020, the EU introduced its European Green Deal investment



plan to mobilize private capital for green bonds, aiming to meet member countries' climate and energy goals. This includes the implementation of a European Green Bond Standard (EuGBS), which requires that at least 85% of the proceeds be aligned with the EU taxonomy for sustainable activities (EC, 2020). Moreover, as part of the post-COVID recovery initiative NextGenerationEU, the European Commission issued its first green bond in October 2021, raising EU€12 billion (bn) (the largest green bond transaction at that time (EC, 2021)). Understanding the impact of these policy initiatives on the green bond market is an important area for future research.

Finally, it is important to consider how these institutional dynamics play out in other central banks with comparable or contrasting policy approaches such as the BoE, or those that have different approaches such as the US Federal Reserve (Fed) or the People's Bank of China (PBoC). The BoE, for example, adopted a climate strategy broadly similar to the ECB to mitigate climate-related financial risks, which it later paused (Best et al., 2025). By contrast, the Fed has deliberately avoided integrating climate considerations into monetary policy, with Chair Jerome Powell (2020) arguing that such choices should be left to elected officials. At the other end of the spectrum, the PBoC is unconstrained by neoliberal norms and has actively offered cheaper capital to banks financing projects that reduce carbon emissions (Best et al., 2025). Comparative and in-depth empirical analyses of these divergent cases are essential to advancing a broader debate about both the practical limits of 'market neutrality' and the constraints facing the recent ideational turn toward green central banking.

## Notes

1. For reference, in 2023 the totality of the Green Climate Fund portfolio totaled to US\$52bn, while the total size of the green bond market was US\$2.8tn (CBI, 2024; Green Climate Fund, 2023).
2. <https://www.bankofengland.co.uk/-/media/boe/files/speech/2015/breaking-the-tragedy-of-the-horizon-climate-change-and-financial-stability.pdf>.
3. Basis point or bps are a widely used unit of measure in finance to show the change in the value or rate of a financial instrument. One basis point is equal to one one-hundredth of a percentage point (0.01%). For example, if the interest rate of an instrument increased by 1%, we could say that it increased by 100 basis points. Using basis points instead of percentages makes it more convenient to compare small changes and variations in financial percentages without introducing the complexity of using decimal points.
4. Given that inclusion in the ECSF carries an 'eligibility premium', in the appendix I test whether this 'eligibility premium' is more important to achieving additionality than the 'green bond premium'.

5. Given that the matching procedure would have controlled for differences in liquidity category as it would match bonds from the same issuer, I diverge from the ECB's approach and further control for liquidity defined as the bid-ask spread. For robustness, I estimate these alternative specifications in the appendix.
6. The full results appear in Table C.1 of the Appendix.
7. The full results appear in Table C.3 in the Appendix.
8. The regression table for this plot appears in Table C.2 of the Appendix.

## Acknowledgments

I am deeply grateful to Neil Carter and Liam Clegg, as well as to Laura Coroneo and the participants of the 11th Asset Pricing Workshop of the Centre for Applied Macro Finance at the University of York for their valuable feedback. I am especially grateful to the editor and the three anonymous reviewers for their insightful comments and constructive feedback, which significantly improved the quality of this paper.

## Disclosure statement

No potential conflict of interest was reported by the author(s).

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*Julio Galindo-Gutiérrez* is a Research Fellow in the Department of Environment and Geography at the University of York working on a project analyzing how capital markets can catalyze sustainable food-system transitions. He has published in *Party Politics* and *Journal of International Relations and Development* and has a forthcoming book on the institutional dynamics of climate finance with Oxford University Press.

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